

Full Molecular Electrolyte Conductivity

This document specifies one computational system that maps an electrolyte recipe, temperature, pressure, and molecular species data to one bulk scalar conductivity. The model resolves the complete periodic molecular liquid. Ion pairs, aggregates, solvent shells, cages, and concentration fluctuations are sampled configurations. They are not configured populations or conductivity corrections.

1 Complete input and output

The physical input is

$$\mathcal{I} = (\mathcal{R}, T, p, \mathcal{S}, \Theta).$$

The recipe $\mathcal{R}$ gives solvent volume ratios, salt molarities, and additive mass fractions. The species data $\mathcal{S}$ give molecular topology, isotope masses, formal molecular charge, interaction sites, and parameters for the common interaction equations. The shared parameter set Θ is fixed over the supported chemical domain.

The numerical arguments are finite-cell scale m , equilibrium integration resolution r , memory resolution ℓ , and basis level n . The complete calculation is

$$\begin{aligned} (\mathcal{R}, T, p, \mathcal{S}, \Theta) &\longmapsto \{N_s^{(m)}\}, V_m^*, q \\ &\longmapsto U_\Theta, \rho_\Theta \\ &\longmapsto K_{\Theta,0}(q), D_\Theta(q), L_\Theta, P \\ &\longmapsto A_n, h_n, C^{\text{dir}} \\ &\longmapsto \sigma_{m,r,\ell,n} \longmapsto \sigma(\mathcal{R}, T, p). \end{aligned}$$

2 Neutral periodic molecular system

Let $x_s(\mathcal{R})$ be the molecular amount of component s after converting the recipe units and expanding salts into molecular ions. At cell scale m , choose nonnegative integers

$$\{N_s^{(m)}\} = \arg \min_{\{n_s\} \subset \mathbb{N}_0} \sum_s w_s \left(\frac{n_s}{m} - x_s(\mathcal{R}) \right)^2$$

subject to

$$\sum_s n_s Q_s = 0, \quad \mathbf{B} \mathbf{n} = \mathbf{b}_m.$$

The first constraint imposes exact molecular electroneutrality. The matrix $\mathbf{B}$ contains exact stoichiometric constraints. The sequence obeys

$$N_s^{(m)} / m \longrightarrow x_s(\mathcal{R}).$$

For periodic cell matrix H , volume $V = \det H$, and constrained molecular configuration space Ω_V , define

$$Z_V(T, \{N_s\}) = \int_{\Omega_V} e^{-\beta U_\Theta(q; H)} \mathrm{d}q, \quad \beta = (k_B T)^{-1}.$$

At pressure p ,

$$\Delta(T, p, \{N_s\}) = \int_0^\infty e^{-\beta p V} Z_V(T, \{N_s\}) \mathrm{d}V,$$

$$V_m^* = \Delta^{-1} \int_0^\infty V e^{-\beta p V} Z_V(T, \{N_s^{(m)}\}) \mathrm{d}V.$$

Density is therefore an output:

$$\rho_{\text{mass}} = \frac{\sum_s N_s^{(m)} M_s}{N_A V_m^*}.$$

3 Molecular energy operator

One interaction form applies to every recipe:

$$\begin{aligned} U_\Theta = & U_{\text{bond}} + U_{\text{angle}} + U_{\text{torsion}} + U_{\text{improper}} \\ & + U_{\text{rep}} + U_{\text{disp}} + U_{\text{elec}} + U_{\text{pol}} + U_{\text{mb}}. \end{aligned}$$

The bonded terms are

$$U_{\text{bond}} = \sum_b \frac{k_b}{2} (r_b - r_b^0)^2, \quad U_{\text{angle}} = \sum_a \frac{k_a}{2} (\theta_a - \theta_a^0)^2,$$

$$U_{\text{torsion}} = \sum_d \sum_k k_{dk} [1 + \cos(n_{dk} \varphi_d - \delta_{dk})],$$

with one declared improper functional form. For the Lennard–Jones realization,

$$U_{\text{rep}} + U_{\text{disp}} = \sum_{i < j}^{\text{nonbonded}} 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right],$$

$$\sigma_{ij} = \frac{\sigma_i + \sigma_j}{2}, \quad \epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}.$$

All periodic electrostatic blocks use the same zero-average Green function:

$$-\nabla^2 G_H(x) = 4\pi \left[\sum_{z \in \mathbb{Z}^3} \delta(x - Hz) - V^{-1} \right].$$

Permanent electrostatics are

$$U_{\text{elec}} = \frac{1}{8\pi\epsilon_0} \sum_{i \neq j} q_i q_j G_H(r_i - r_j) + U_{\text{self}} + U_{\text{excl}}.$$

The exclusion term applies bonded exclusions and 1–4 scaling to the same periodic Green function, including its reciprocal-space correction.

For induced dipoles μ , define the periodic field $E_H(q)$ and interaction matrix $T_H(q)$ from derivatives of G_H :

$$U_{\text{pol}}(q, \mu) = \frac{1}{2} \mu^\top \alpha^{-1} \mu - \mu^\top E_H(q) - \frac{1}{2} \mu^\top T_H(q) \mu.$$

The same declared short-range damping applies to charge–dipole and dipole–dipole blocks. The physical induced dipoles and energy are

$$\mu^*(q) = \arg \min_\mu U_{\text{pol}}(q, \mu), \quad U_{\text{pol}}(q) = U_{\text{pol}}(q, \mu^*(q)).$$

The many-body term U_{mb} is a declared permutation-, rotation-, and translation-invariant function of atomic geometry. Forces and virial derive from this single energy:

$$F_i = -\nabla_{r_i} U_\Theta, \quad \Xi = -\frac{\partial U_\Theta}{\partial H} H^\top.$$

4 Equilibrium measure

At V_m^* ,

$$\rho_\Theta(q) \mathrm{d}q = Z^{-1} e^{-\beta U_\Theta(q; V_m^*)} \mathrm{d}q.$$

Rigid constraints $g(q) = 0$ restrict this measure to the constraint manifold with its induced volume element. Equilibrium integration uses a sequence $\mathcal{Q}_r$ with

$$\mathcal{Q}_r[f] \longrightarrow \langle f \rangle_{\rho_\Theta} = \int f(q) \rho_\Theta(q) \mathrm{d}q.$$

Pair basins, coordination shells, aggregates, cages, and ligand states are smooth observables of q . They do not alter ρ_Θ .

5 Molecular coordinates and null modes

For molecule a ,

$$R_a = \frac{1}{M_a} \sum_{i \in a} m_i r_i, \quad V_a = \frac{1}{M_a} \sum_{i \in a} m_i v_i, \quad F_a = \sum_{i \in a} F_i.$$

Collect the molecular translational and rotational velocities in V , with mass–inertia matrix M . Let the columns of B span uniform translation and exact rigid-constraint null modes. The physical-range projector is

$$\Pi = I - B(B^\top B)^\# B^\top.$$

Memory, friction, diffusion, and Poisson operators act on $\text{ran } \Pi$.

6 Microscopic dynamics for molecular memory

The memory kernel is generated before the configuration diffusion and therefore cannot be evolved by L_Θ . Introduce constrained atomic momenta p , mass matrix M_{at} , and the phase-space Hamiltonian

$$\mathcal{H}_\Theta(q, p) = \frac{1}{2} p^\top M_{\text{at}}^{-1} p + U_\Theta(q).$$

Write $G(q) = \nabla_q g(q)$, $v = M_{\text{at}}^{-1} p$, and $A_g(q) = G(q) M_{\text{at}}^{-1} G(q)^\top$. On the cotangent bundle of the molecular constraint manifold,

$$\dot{q} = v, \quad \dot{p} = -\nabla_q U_\Theta - G(q)^\top \lambda,$$

where the acceleration constraint determines the multiplier without an additional physical parameter:

$$\lambda = A_g(q)^\# \left[\dot{G}(q; v) v - G(q) M_{\text{at}}^{-1} \nabla_q U_\Theta \right].$$

The microscopic generator is therefore

$$\mathcal{G}_\Theta f = (M_{\text{at}}^{-1} p)^\top \nabla_q f - \left(\nabla_q U_\Theta + G^\top \lambda \right)^\top \nabla_p f,$$

on $g(q) = 0$ and $G(q) M_{\text{at}}^{-1} p = 0$. Its invariant canonical measure is

$$\varrho_\Theta(q, p) \mathrm{d}q \mathrm{d}p = \mathcal{Z}^{-1} e^{-\beta \mathcal{H}_\Theta(q, p)} \mathrm{d}\lambda_g(q) \mathrm{d}\lambda_{T_q}(p).$$

Here λ_g and λ_{T_q} are the induced configuration and tangent momentum measures. This phase-space dynamics uses exactly the energy, masses, cell, and constraints that define ρ_Θ . It is used to evaluate molecular memory; it is not the later homogenized configuration process.

7 Conditional memory and diffusion

Let $\mathcal{P}$ be conditional expectation in $L^2(\varrho_\Theta)$ onto the resolved molecular centers, orientations, and their conjugate translational and rotational momenta. Set $\mathcal{Q} = I - \mathcal{P}$. The orthogonal molecular force and its orthogonal evolution are

$$F^\perp = \mathcal{Q}F, \quad F^\perp(t) = e^{t\mathcal{Q}\mathcal{G}_\Theta}\mathcal{Q}F^\perp.$$

The fluctuation–dissipation definition of the conditional memory kernel is

$$K_\Theta(q, t) = \beta \mathbb{E}_{\varrho_\Theta} \left[\Pi F^\perp(t) F^\perp(0)^\top \Pi \mid q(0) = q \right].$$

The conditional expectation integrates canonical momenta and unresolved molecular coordinates at fixed resolved configuration. Thus K_Θ is determined by $(U_\Theta, M_{\text{at}}, \varrho_\Theta, \mathcal{P})$, independently of D_Θ and L_Θ . Define

$$C_{VV}(q, t) = \mathbb{E}[\Pi V(t) V(0)^\top \Pi \mid q(0) = q], \quad C_{PV}(q, t) = M C_{VV}(q, t).$$

These obey the matrix Volterra equation

$$\frac{\partial C_{PV}(q, t)}{\partial t} = - \int_0^t K_\Theta(q, \tau) C_{VV}(q, t - \tau) d\tau.$$

For $s > 0$,

$$\tilde{K}_\Theta(q, s) = \left[C_{PV}(q, 0) - s \tilde{C}_{PV}(q, s) \right] \tilde{C}_{VV}(q, s)^\#.$$

The zero-frequency friction and molecular diffusion are

$$\begin{aligned} K_{\Theta,0}(q) &= \lim_{s \downarrow 0} \Pi \tilde{K}_\Theta(q, s) \Pi, \\ D_\Theta(q) &= k_B T \Pi K_{\Theta,0}(q)^\# \Pi. \end{aligned}$$

For finite trajectory resolution ℓ , evaluate the same zero-frequency operator through the positive covariance-growth form

$$D_{\Theta,\ell}(q) = \frac{1}{2t_\ell} \mathbb{E} \left[\Delta R_\ell \Delta R_\ell^\top \mid q(0) = q \right], \quad K_{\Theta,0,\ell} = k_B T D_{\Theta,\ell}^\#.$$

where $\Delta R_\ell = \int_0^{t_\ell} \Pi V(t) dt$ and the conditional mean displacement is removed. This form is positive semidefinite at every finite resolution and converges to the same zero-frequency mobility as the Laplace–Volterra expression when the velocity correlation is integrable. It avoids replacing a noisy indefinite kernel estimate by clipped eigenvalues. The limit is evaluated on the physical range. Conserved and explicitly resolved slow coordinates remain outside the inversion.

The replacement of the non-Markovian molecular equation by $D_\Theta(q)$ is its zero-frequency homogenization. When the orthogonal-force kernel is integrable and the eliminated variables mix on a finite time scale, this homogenized operator has the same zero-frequency molecular mobility as the generalized Langevin equation.

8 Constrained reversible generator

Let T_q project Cartesian molecular increments onto the tangent space of the rigid constraints. Replace D_Θ by $T_q D_\Theta T_q^\top$ and retain the same symbol. The production generator is

$$L_\Theta f = \rho_\Theta^{-1} \nabla_q \cdot (\rho_\Theta D_\Theta \nabla_q f).$$

In coordinates,

$$L_\Theta f = -\beta \nabla U_\Theta^\top D_\Theta \nabla f + (\nabla \cdot D_\Theta)^\top \nabla f + D_\Theta : \nabla^2 f.$$

The associated stochastic dynamics is

$$dq_t = [-\beta D_\Theta \nabla U_\Theta + \nabla \cdot D_\Theta] dt + \sqrt{2D_\Theta} dW_t.$$

The divergence drift and correlated noise belong to the same operator. For smooth constraint-compatible functions,

$$-\langle f, L_\Theta g \rangle_{\rho_\Theta} = \left\langle \nabla f^\top D_\Theta \nabla g \right\rangle_{\rho_\Theta}.$$

Thus L_Θ is reversible and negative semidefinite on mean-zero functions.

9 Charge polarization

Use formal molecular charges and unwrapped molecular centers:

$$P(q) = \sum_a Q_a R_a^{\text{unwrap}}.$$

Molecular translation carries unbounded charge displacement. Intramolecular rotation, vibration, and induced polarization are bounded unless molecular charge topology changes. The drift current is

$$j_i(q) = L_\Theta P_i(q).$$

The stochastic quadratic variation supplies

$$C_{ij}^{\text{dir}} = \left\langle \nabla P_i^\top D_\Theta \nabla P_j \right\rangle_{\rho_\Theta}.$$

10 Poisson corrector and scalar conductivity

For each Cartesian component solve

$$-L_\Theta \chi_i = L_\Theta P_i, \quad \langle \chi_i \rangle_{\rho_\Theta} = 0.$$

The weak equation is

$$\left\langle \nabla \varphi^\top D_\Theta \nabla \chi_i \right\rangle = - \left\langle \nabla \varphi^\top D_\Theta \nabla P_i \right\rangle$$

for every admissible mean-zero test function φ . The homogenized charge diffusion is

$$\mathcal{D}_{ij}^Q = C_{ij}^{\text{dir}} - \left\langle L_\Theta P_i, (-L_\Theta)^\# L_\Theta P_j \right\rangle_{\rho_\Theta}.$$

Equivalently,

$$\mathcal{D}_{ii}^Q = \min_{\chi \in \mathcal{H}_E} \left\langle (\nabla P_i + \nabla \chi)^\top D_\Theta (\nabla P_i + \nabla \chi) \right\rangle_{\rho_\Theta}.$$

With P measured in C m ,

$$\sigma(\mathcal{R}, T, p) = \frac{1}{3k_B T V_m^*} \text{tr } \mathcal{D}^Q \quad [\text{S m}^{-1}].$$

11 Constructive nested basis

Generate smooth permutation-, translation-, and rotation-compatible functions from radial atomic densities, angular invariants, molecular orientations, coordination numbers, charge-density Fourier modes, concentration modes, cluster graphs, and hierarchical products. Let $\mathcal{G}_n$ include radial and angular order, reciprocal wavevector norm, correlation order, and cluster depth at most n . After centering and removal of exact null functions,

$$V_n = \text{span } \mathcal{G}_n, \quad V_n \subset V_{n+1}.$$

The hierarchy obeys

$$\bigcup_{n=1}^{\infty} V_n = \mathcal{H}_E, \quad \|f\|_E^2 = \langle \nabla f^\top D_\Theta \nabla f \rangle.$$

For basis $\Phi_n = (\phi_1, \dots, \phi_{d_n})$, compute

$$A_{\mu\nu}^{(n)} = \left\langle \nabla \phi_\mu^\top D_\Theta \nabla \phi_\nu \right\rangle,$$

$$h_{\mu i}^{(n)} = \left\langle \nabla \phi_\mu^\top D_\Theta \nabla P_i \right\rangle.$$

The coefficient vector solves

$$A^{(n)} c_i^{(n)} = -h_i^{(n)}.$$

The finite charge diffusion and conductivity are

$$\mathcal{D}_{ij}^{Q,n} = C_{ij}^{\text{dir}} - (h_i^{(n)})^\top (A^{(n)})^\# h_j^{(n)},$$

$$\sigma_{m,r,\ell,n} = \frac{\text{tr}[C^{\text{dir}} - h_n^\top A_n^\# h_n]}{3k_B T V_m^*}.$$

12 Residual-driven enrichment

For $g \in V_{n+1} \setminus V_n$, define

$$\tilde{A}_{gg} = A_{gg} - A_{g\Phi} A_{\Phi\Phi}^\# A_{\Phi g},$$

$$\tilde{h}_{g,i} = h_{g,i} - A_{g\Phi} A_{\Phi\Phi}^\# h_{\Phi,i}.$$

When $\tilde{A}_{gg} > 0$, the exact gain in the variational correction is

$$S(g) = \sum_{i=1}^3 \frac{|\tilde{h}_{g,i}|^2}{\tilde{A}_{gg}}.$$

Candidates come from the next declared hierarchy level. Exhausting one finite dictionary does not establish convergence.

13 Galerkin convergence

Let χ_i^* solve the full Poisson problem and $\chi_{n,i}$ be its Galerkin projection. Galerkin orthogonality gives

$$\left\langle \nabla (\chi_i^* - \chi_{n,i})^\top D_\Theta \nabla \phi \right\rangle = 0, \quad \phi \in V_n.$$

Expanding the variational energy gives

$$\sigma_{m,r,\ell,n} - \sigma_{m,r,\ell,\infty} = \frac{1}{3k_B T V_m^*} \sum_{i=1}^3 \|\chi_i^* - \chi_{n,i}\|_E^2 \geq 0.$$

Therefore

$$\sigma_{m,r,\ell,n} \downarrow \sigma_{m,r,\ell,\infty}.$$

When equilibrium integration converges, the zero-frequency memory limit exists on the physical range, the basis hierarchy is energy-norm dense, and the thermodynamic limit exists,

$$\sigma(\mathcal{R}, T, p) = \lim_{m \rightarrow \infty} \lim_{r \rightarrow \infty} \lim_{\ell \rightarrow \infty} \lim_{n \rightarrow \infty} \sigma_{m,r,\ell,n}.$$

Every limit refines a named numerical approximation while $U_\Theta, K_\Theta, P, T, p$, and the intensive recipe remain fixed.

14 Executable calculation order

1. Decode $\mathcal{R}$ and construct neutral integer counts.
2. Evaluate U_{Θ} and equilibrate at (T, p) to obtain V_m^* and samples of ρ_{Θ} .
3. Construct $\mathcal{G}_{\Theta}$ from the same energy, masses, cell, and molecular constraints; sample its canonical phase-space measure.
4. Project atomic coordinates, momenta, velocities, and forces to constrained molecular variables and form $\mathcal{P}$ and $\mathcal{Q}$.
5. Evolve orthogonal forces with $e^{tQ\mathcal{G}_{\Theta}Q}$, evaluate their conditional correlations, and solve for $K_{\Theta,0}(q)$ on the physical range.
6. Form $D_{\Theta}(q) = k_B T K_{\Theta,0}(q)^{\#}$.
7. Evaluate C^{dir} , A_n , and h_n over the same equilibrium measure.
8. Solve $A_n c_{n,i} = -h_{n,i}$ after removing exact null modes.

9. Return

$$\sigma_{m,r,\ell,n} = \frac{\text{tr}(C^{\text{dir}} - h_n^{\top} A_n^{\#} h_n)}{3k_B T V_m^*}.$$

10. Refine (m, r, ℓ, n) without changing the physical model.

15 Model boundary

The equations contain no configured SSIP or CIP fractions, mass-action population allocation, species conductivity bonus, terminal conductivity multiplier, or recipe-specific transport branch. Species data determine molecular construction and interaction parameters. The equilibrium measure determines structure. Conditional memory determines diffusion. The Poisson corrector determines correlated charge transport. Their composition returns the scalar conductivity.